

# Max Flow 4

## Max-Flow Min-Cut Algorithms

1. Ford-Fulkerson
  - Find augmenting paths using DFS or BFS
  - $O(mC)$  time where  $C$  = size of max flow
    - Assumes integer capacities
2. Edmonds-Karp
  - Find augmenting paths using BFS
  - $O(m^2n)$  time
    - No assumptions on capacities

## Ford-Fulkerson Algorithm

1. Input: Flow network  $G=(V,E)$  with positive integer capacities  $c_e$ 
  - Set  $f_e=0$  for all  $e$  in  $E$
  - Build residual network  $G^f$
  - Check for st-path  $P$  in  $G^f$  using BFS or DFS
  - If no such path, return( $f$ )
  - Let  $c(P) = \min$  capacity along  $P$  in  $G^f$
  - Augment  $f$  by  $c(P)$  units along  $P$
  - Go to step 2

## Edmonds-Karp Algorithm

1. Input: Flow network  $G=(V,E)$  with positive capacities  $c_e$ 
  - Set  $f_e=0$  for all  $e$  in  $E$
  - Build residual network  $G^f$
  - Check for st-path  $P$  in  $G^f$  using BFS
  - If no such path, return( $f$ )
  - Let  $c(P) = \min$  capacity along  $P$  in  $G^f$
  - Augment  $f$  by  $c(P)$  units along  $P$
  - Go to step 2

## Proof Outline

1. Running time:  $O(m^2n)$ 
  - Number of rounds  $\leq mn$
2. In every round, residual graph changes ( $\geq 1$  edge deleted)
3. Key lemma: For every edge  $e$ ,  $e$  is deleted and reinserted later  $\leq n/2$  times
  - Since  $m$  edges are in the graph,  $\leq mn/2$  total rounds

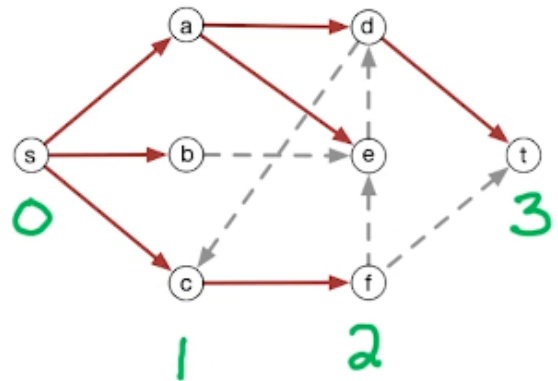
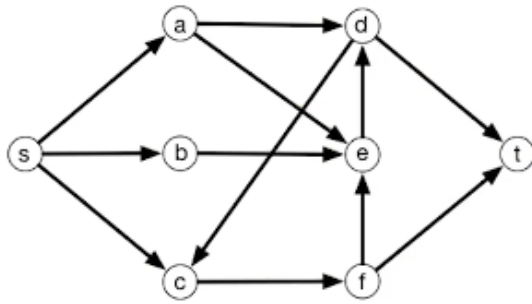
## BFS Properties

1. Lemma: For every edge  $e$ ,  $e$  is deleted and reinserted later  $\leq n/2$  times
2. BFS
  - Input: directed  $G=(V,E)$ , no edge weights,  $s$  in  $V$
  - Output: For all  $v$  in  $V$ ,  $\text{dist}(v) = \min$  number of edges  $s \rightarrow v$

## BFS Example

1. Run BFS on the following example
  - $\text{level}(v) = \text{dist}(v) = \min$  number of edges  $s \rightarrow v$
  - $P = s \rightarrow a \rightarrow d \rightarrow t$

- Levels:  $s = 0$ ,  $a = 1$ ,  $d = 2$ ,  $t = 3$



BFS Example

## BFS Properties - Part 2

- How does  $\text{level}(z)$  change as  $G^f$  changes?
  - Claim: For every  $z$  in  $V$ ,  $\text{level}(z)$  does not decrease

## Add/Delete Edges

- How does  $G^f$  change in a round?
  - For  $vw$  in  $E$ 
    - Add  $vw$  if flow was full and then reduced
      - \* This means  $wv$  is in  $P$
    - Remove  $vw$  if flow is now full
      - \* This means  $vw$  is in  $P$
    - Add  $wv$  if flow was empty
      - \* This means  $vw$  is in  $P$
    - Remove  $wv$  if flow was positive and now empty
      - \* This means  $wv$  is in  $P$

## Conclusion

- If add  $yz$  to  $G^f$  then  $zy$  is in  $P$
- If remove  $yz$  to  $G^f$  then  $yz$  is in  $P$

## Proof of Claim (Edmonds-Karp Algorithm)

- Claim: For every vertex  $z$  in  $V$ ,  $\text{level}(z)$  does not decrease
  - Might decrease if we add edge  $yz$ 
    - Suppose  $\text{level}(z) = i$
    - Add  $yz$  to  $G^f$ , so  $zy$  in  $P$
    - Because  $P$  is a BFS path,  $\text{level}(y) = \text{level}(z) + 1$ 
      - \* This means  $\text{level}(y) = i + 1$
      - \* Level does not decrease

## Delete/Add Effect

- Say  $\text{level}(v) = i$ 
  - Suppose we delete  $vw$  from  $G^f$  so  $vw$  is on the augmented path  $P$

- $\text{level}(w) = \text{level}(v) + 1 \geq i + 1$
- Later add  $vw$  in  $G^f$  so  $wv$  in  $P$ 
  - $\text{level}(v) = \text{level}(w) + 1 \geq i + 2$
- So  $\text{level}(v)$  increases by  $\geq 2$

## Finishing Off

1. If we delete  $vw$  from  $G^f$  and later add  $vw$ , then  $\text{level}(v)$  increases by  $\geq 2$
2. Minimum level = 0, Maximum level =  $n$ 
  - delete + add  $vw \leq n/2$  times meaning  $\leq nm$  rounds